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Evaluation (of Classification and Clustering) and Parameterization

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Agenda

Multiclass Classification Evaluation

- Every classification process is subject to errors
- It is necessary to use statistical procedures capable of defining the accuracy of the results
- ...only then can we judge whether they are reliable or not

Confusion Matrix

- One of the most efficient ways to represent classification accuracy
- Capable of describing both inclusion and exclusion errors
- Inclusion — a pattern is assigned to an incorrect class
- Exclusion — a pattern is not assigned to its expected class

Tabela: Elements of the confusion matrix.

		Reference				
		ω_1	ω_2	$\cdots$	ω_c	
Classification	ω_1	a_{11}	a_{12}	$\cdots$	a_{1c}	a_{1+}
	ω_2	a_{21}	a_{22}	$\cdots$	a_{2c}	a_{2+}
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	ω_c	a_{c1}	a_{c2}	$\cdots$	a_{cc}	a_{c+}
		a_{+1}	a_{+2}	$\cdots$	a_{+c}	m

Accuracy Measures

Overall Accuracy

A simple measure that expresses the ratio between the total number of correct predictions and the total number of observations considered, that is:

$$A_g = \frac{1}{m} \sum_{i=1}^{c} a_{ii} \tag{1}$$

Producer's and User's Accuracy

- Producer — the probability that an observation has been correctly classified
- User — the probability that the observation actually represents the assigned class

For ω_i, the corresponding producer's and user's accuracies are:

$$A_{pi} = \frac{a_{ii}}{a_{+i}} \qquad A_{ui} = \frac{a_{ii}}{a_{i+}} \tag{2}$$

Accuracy Measures (cont.1)

Kappa Coefficient

- Widely used in classification evaluation
- Measures the agreement between the predicted classification and the actual reference
- Can be used to determine whether the achieved accuracy is significantly better than a random classification

$$\kappa = \frac{A_g - A_a}{1 - A_a}, \quad \kappa \in] - \infty, 1] \tag{3}$$

where A_g corresponds to the overall accuracy and $A_a = \frac{\sum_{i=1}^{c} a_{i+} a_{+i}}{m^2}$ refers to the accuracy expected by chance

Accuracy Measures (cont.2)

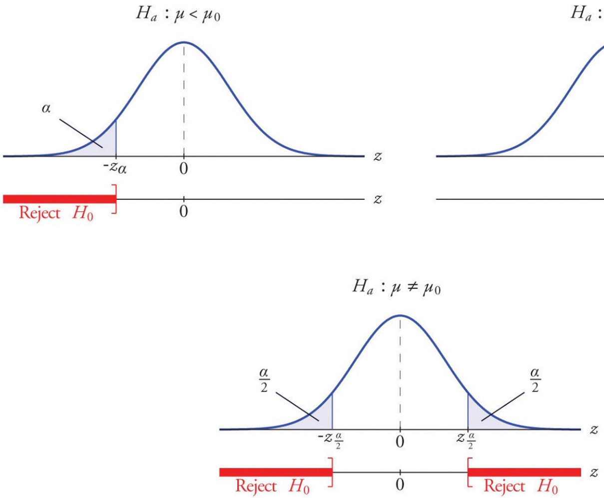
Variance of the Kappa Coefficient

$$\sigma_\kappa^2 = \frac{1}{m}\left(\frac{\phi_1(1-\phi_1)}{(1-\phi_1)^2} + \frac{2(1-\phi_1)(2\phi_1\phi_2-\phi_3)}{(1-\phi_2)^3} + \frac{(1-\phi_1)^2(\phi_4-4\phi_2)^2}{(1-\phi_2)^4}\right), \quad (4)$$

where $\phi_1 = A_g$; $\phi_2 = A_a$;

$\phi_3 = \frac{1}{m^2}\sum_{i=1}^{c} a_{ii}(a_{i+} + a_{+i})$; $\phi_4 = \frac{1}{m^3}\sum_{i=1}^{c}\sum_{j=1}^{c} a_{ij}(a_{j+} + a_{+i})$.

- Considering the coefficients κ_1 and κ_2, with respective variances $\sigma_{\kappa_1}^2$ and $\sigma_{\kappa_2}^2$, it is known that the statistic $z_\kappa = \frac{\kappa_1 - \kappa_2}{\sqrt{\sigma_{\kappa_1}^2 + \sigma_{\kappa_2}^2}} \sim \mathcal{N}(0,1)$
- Under these conditions, it is possible to state that the classifications associated with κ_1 and κ_2 are statistically significant at level α when the probability of obtaining a value equal to or more extreme than z_κ is less than $1-\alpha$.
- This discussion summarizes a hypothesis testing procedure


$H_a : \mu < \mu_0$
α
$-z_\alpha$
0
z
Reject H_0
$H_a : \mu > \mu_0$
z_α
Reject H_0
$H_a : \mu \neq \mu_0$
$\frac{\alpha}{2}$
$-z_{\frac{\alpha}{2}}$
$z_{\frac{\alpha}{2}}$
Reject H_0
Reject H_0

Accuracy Measures (cont.3)

Tau Coefficient

- For the kappa coefficient, chance agreement is quantified by A_a, and its contribution is subsequently removed from the overall accuracy
- The estimation of these random agreements is based on the number of observations distributed across each class with respect to the data used in the assessment
- Assuming that random agreements occur uniformly across the c classes, we obtain the tau agreement coefficient:

$$\tau = \frac{A_g - \frac{1}{c}}{1 - \frac{1}{c}} \quad \tau \in [-1, 1] \tag{5}$$

Accuracy Measures (cont.4)

Variance of the Tau Coefficient

$$\sigma_\tau^2 = \frac{1}{m}\left[\frac{A_g(1-A_g)}{\left(1-\frac{1}{c}\right)^2}\right] \tag{6}$$

- The statistic $z_\tau = \dfrac{\tau_1 - \tau_2}{\sqrt{\sigma_{\tau_1}^2 + \sigma_{\tau_2}^2}} \sim \mathcal{N}(0,1)$ can be used to verify the significance between two classifications with coefficients τ_1 and τ_2

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Binary Classifications

- In specific situations, the classification problem is reduced to only two classes: one indicating the presence/occurrence of a given condition, and the other indicating its absence
- *TP* and *TN* represent the number of correctly predicted cases of occurrence and non-occurrence of the condition, respectively
- *FP* and *FN* represent incorrect predictions, where *FP* is the number of positive predictions that should have been negative, and *FN* is the number of negative predictions that should have been positive

		Reference	
		Positive Condition	Negative Condition
Prediction	Condition Positive	True Positive (*TP*)	False Positive (*FP*)
	Condition Negative	False Negative (*FN*)	True Negative (*TN*)

F_β Measure

- It consists of a harmonic mean involving two other indices:
  - Precision: $Pr = \dfrac{TP}{TP + FP}$
  - Recall: $Re = \dfrac{TP}{TP + FN}$.
- Once precision and recall are defined, we have:

$$F_\beta = (1 + \beta^2)\frac{Pr \cdot Re}{(\beta^2 \cdot Pr) + Re} \tag{7}$$

- Some typical values for β are 1, 2, and $1/2$:
  - $\beta = 1$ – the measure is reduced to the *F1-Score* or *F-Measure*
  - $\beta = 2$ – gives more importance to recall
  - $\beta = 1/2$ – gives more importance to precision, thus reducing the influence of false negatives
- F_β does not account for true negatives, which may lead to misleading interpretations when the number of observations used to evaluate the results is not equally distributed between the two classes involved in the problem

Matthews Correlation Coefficient

- Similar in use to F_β, especially when the number of observations used for evaluation differs between classes

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \tag{8}$$

- MCC values are limited to $[-1, 1]$
- -1 and 1 indicate completely disagreeing and perfectly agreeing predictions, respectively
- Values close to 0 indicate agreement close to random chance

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Cluster Evaluation

- The evaluation process can differ between measures based on reference observations or fully autonomous measures
- The *V*-measure requires reference data for its calculation
- The silhouette coefficient (*SC*) and the Calinski-Harabasz index (*CHI*) provide ways to assess the observed partitioning based on statistics extracted directly from the clusters

V-measure

- Let $\mathcal{I}$ be a set of patterns partitioned into clusters $\mathcal{G}_1, \ldots, \mathcal{G}_k$
- From a known reference dataset $\mathcal{D}$, we know that elements of $\mathcal{I}$ are associated with one of the classes in $\Omega = \{\omega_1, \ldots, \omega_c\}$
- In $\mathcal{D}$ there are m_j examples associated with class ω_j, such that $\sum_{j=1}^{c} m_j = m$

V-measure

- Evaluates the partitioning of $\mathcal{I}$ based on:
  - Homogeneity – $H_o = 1 - \frac{h_{c|k}}{h_c}$ … expects that identified clusters are composed of examples from a single class
  - Completeness – $C_o = 1 - \frac{h_{k|c}}{h_k}$ … ensures that examples from the same class are assigned to the same cluster
- These characteristics are based on entropy measures:

$$h_c = -\sum_{j=1}^{c} \frac{m_j}{m} \cdot \log\left(\frac{m_j}{m}\right), \tag{9}$$

$$h_{c|k} = -\sum_{j=1}^{c}\sum_{\ell=1}^{k} \frac{m_{j\ell}}{m} \cdot \log\left(\frac{m_{j\ell}}{m_\ell}\right), \tag{10}$$

- h_c refers to the class entropy
- $h_{c|k}$ refers to the class entropy conditioned on the clusters

V-measure (cont.1)

- Finally, the V-measure is defined as:

$$V = 2\frac{H_o \cdot C_o}{H_o + C_o}. \tag{11}$$

- The values returned by the V-measure are within the range $[0, 1]$, where 1 indicates a perfect match between the clusters and the classes defined according to the reference information
- On the other hand, for random clusters, this measure may lead to different values for H_o and C_o, both different from zero

SC-measure

Assuming that a set of patterns $\mathcal{I}$ is partitioned into clusters $\mathcal{G}_1, \ldots, \mathcal{G}_k$, the SC measure is computed as:

$$SC = \frac{S_E - S_I}{\max\{S_I, S_E\}}, \tag{12}$$

where:

$$S_I = \frac{2}{k} \sum_{j=1}^{k} \frac{1}{m_j(m_j - 1)} \sum_{\substack{\mathbf{x}_i, \mathbf{x}_\ell \in \mathcal{G}_j \\ \mathbf{x}_i \neq \mathbf{x}_\ell}} \|\mathbf{x}_i - \mathbf{x}_\ell\| \tag{13}$$

$$S_E = \frac{1}{k} \sum_{j=1}^{k} \frac{1}{m_j m_{p_j}} \sum_{\mathbf{x}_i \in \mathcal{G}_j} \sum_{\mathbf{x}_\ell \in \mathcal{G}_{p_j}} \|\mathbf{x}_i - \mathbf{x}_\ell\|; \;\; p_j = \underset{\substack{p_j = 1, \ldots, k \\ p_j \neq j}}{\arg\max} \|\mu_j - \mu_{p_j}\| \tag{14}$$

where μ_j and μ_{p_j} are the mean vectors (i.e., centroids) representing clusters $\mathcal{G}_j$ and $\mathcal{G}_{p_j}$, respectively, for $j \neq p_j \in \{1, \ldots, k\}$.

CHI-measure

- The *CHI* formulation is based on clustering criteria used in sum-of-squares-based methods
- It is given by:

$$CHI = \frac{Tr(\mathbf{V}_E)}{Tr(\mathbf{V}_I)} \cdot \frac{\#\mathcal{I} - k}{k - 1}, \tag{15}$$

  where $\mathbf{V}_I$ and $\mathbf{V}_E$ quantify the within-cluster variability and the separation between clusters, respectively
- The values of this measure tend to increase as the clusters become denser and more separable from each other
- However, due to the presence of $\mathbf{V}_I$ in its formulation, the *CHI* measure tends to decrease when the identified clusters are not compact

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Parameterization Strategies

- Prior to the training stage of classification and clustering methods, it is common to select hyperparameters associated with the method
- The selection of hyperparameters is the responsibility of the user
- If carried out inappropriately, it may interfere with the method's training and negatively impact the accuracy of the results
- *Grid Search* and *Randomized Search* procedures can assist in hyperparameter selection

Grid Search

- Consists of exhaustively evaluating the performance/accuracy of different parameter combinations
- Supports the selection of a suitable configuration — one that maximizes performance
- The accuracy values observed for the different parameter configurations are computed through the classification/clustering of part of the patterns from the training set, while the remaining patterns of this set are used for the training process
- Typically, the v-fold cross-validation procedure is employed to compute the accuracy associated with each parameter configuration, thus avoiding overfitting

Grid Search + Cross-Validation

This procedure consists of:

1. Partitioning the set $\mathcal{D}$ into v mutually exclusive subsets with approximately the same cardinality
2. Selecting one subset for method evaluation, training the model with the union of the remaining $v - 1$ subsets
3. After repeating these steps for each of the v subsets, a descriptive measure (e.g., the mean) of the observed accuracy values is computed at the end

Randomized Search

- The exhaustive nature of *Grid Search*, especially when the number of cross-validations (v) increases, makes the procedure computationally expensive
- *Randomized Search* emerges as an alternative
- A fixed number of parameter configurations are randomly selected (according to a distribution) and evaluated, whether or not they are subjected to multiple cross-validations

Grid Search and Randomized Search

```
# Libraries and functions used
import scipy.stats
from sklearn import metrics
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import RandomizedSearchCV

# Definition of the search space for Grid Search
searchSpaceGS = {'kernel': ['linear', 'rbf'],
                 'C':[0.1,1,10,100],
                 'gamma':[0.001,0.01,0.1,1.0,1.5,2.0,5.0],
                 'decision_function_shape':['ovo','ovr']}

# Instantiating the procedure
GS = GridSearchCV(svm.SVC(), searchSpaceGS, cv=10,
                  scoring=metrics.make_scorer(
                  metrics.cohen_kappa_score))

# Executing the procedure
GS.fit(xD,yD)
```

Listing: Application of the *Grid Search* and *Randomized Search* procedures.

Grid Search and Randomized Search (cont.1)

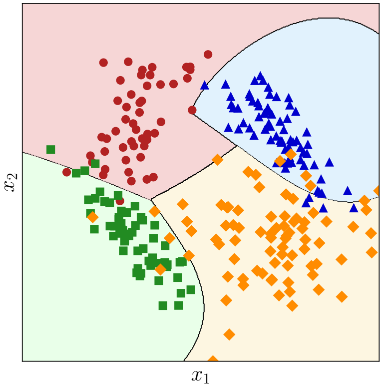
```python
# Checking the results
print('--- Grid Search ---')
print('Best parameters: ',GS.best_params_)
print('Accuracy of the best parameters: ', GS.best_score_)

# Definition of the search space for Randomized Search
searchSpaceRS = {'kernel':['linear','rbf'],
                 'C':scipy.stats.loguniform(0.1,100),
                 'gamma':scipy.stats.uniform(0.001,5),
                 'decision_function_shape':['ovo','ovr']}

# Instantiating the procedure
RS = RandomizedSearchCV(svm.SVC(), searchSpaceRS,
                        scoring = metrics.make_scorer(
                        metrics.cohen_kappa_score),
                        cv=10, n_iter=20)

# Executing the procedure
RS.fit(xD,yD)

# Checking the results
print('\n--- Randomized Search ---')
print('Best configuration: ',RS.best_params_)
print('Observed accuracy: ',RS.best_score_)
```

Grid Search and Randomized Search (cont.2)

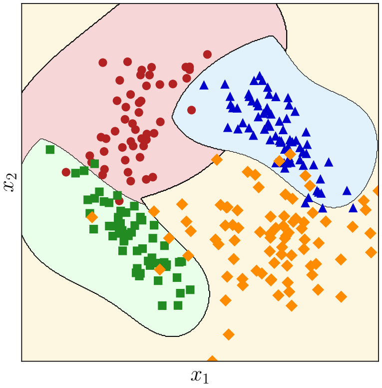
```
--- Grid Search ---
Best parameters:  {'C': 10, 'decision_function_shape': 'ovo',
                   'gamma': 1.5, 'kernel': 'rbf'}
Accuracy of the best parameters:  0.905868705157212
```

```
--- Randomized Search ---
Best configuration:  {'C': 34.24942525733109,
                      'decision_function_shape': 'ovr',
                      'gamma': 3.993972978825609 8,
                      'kernel': 'rbf'}
Observed accuracy:  0.9054692268306892
```

Randomized Search



(a) *Grid Search*



(b) *Randomized Search*

Figura: Classification results obtained using the SVM method after parameterization suggested by *Grid Search* and *Randomized Search* procedures.

Evaluation (of Classification and Clustering) and Parameterization

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